

Lesson 5:

Using R to visualize data



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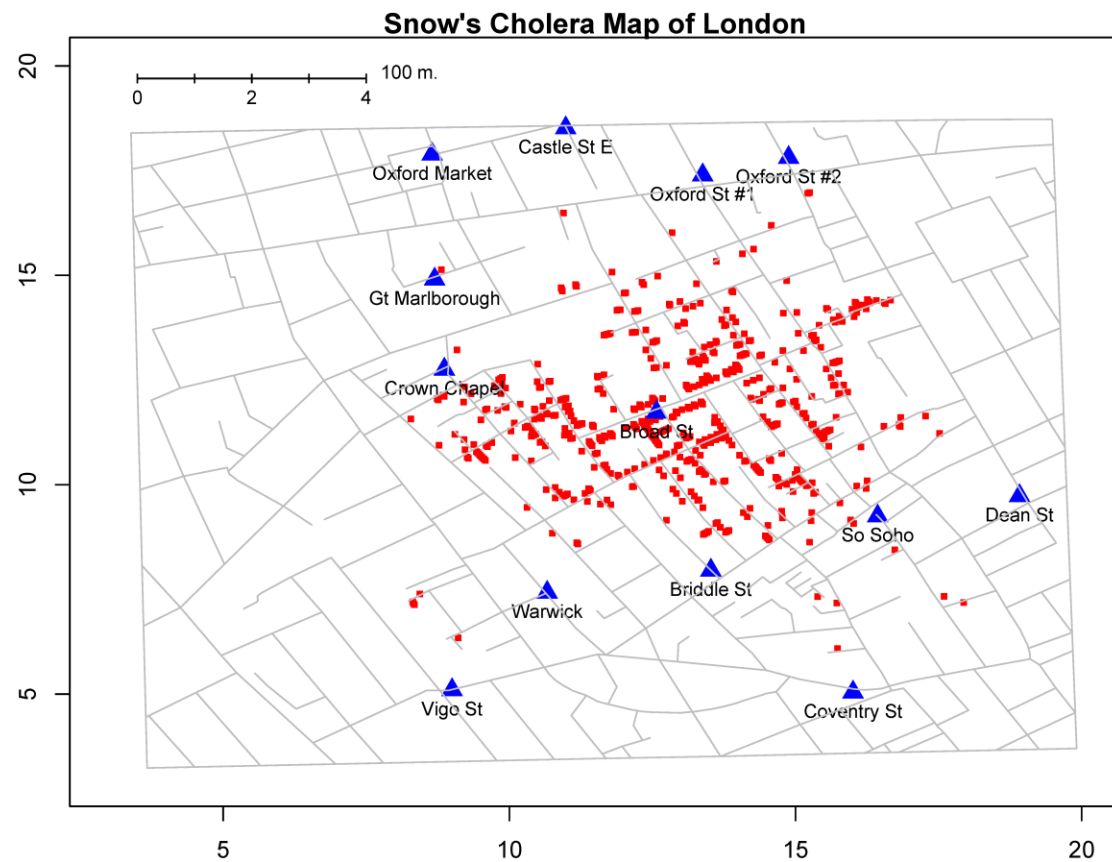
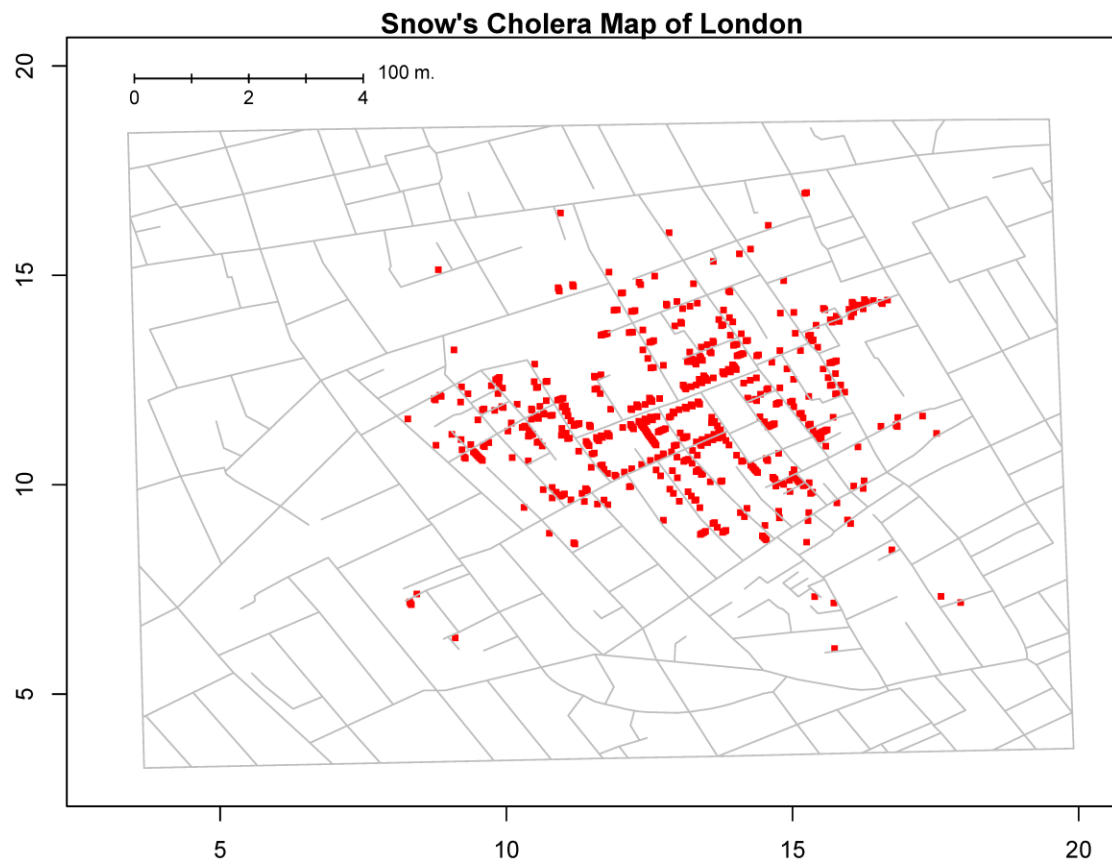
Outlines

- The website: <https://chenziqu0506-ai.github.io/ziqichenRsimpletutorial/lesson5.html>
- Materials also could be downloaded via the above link:

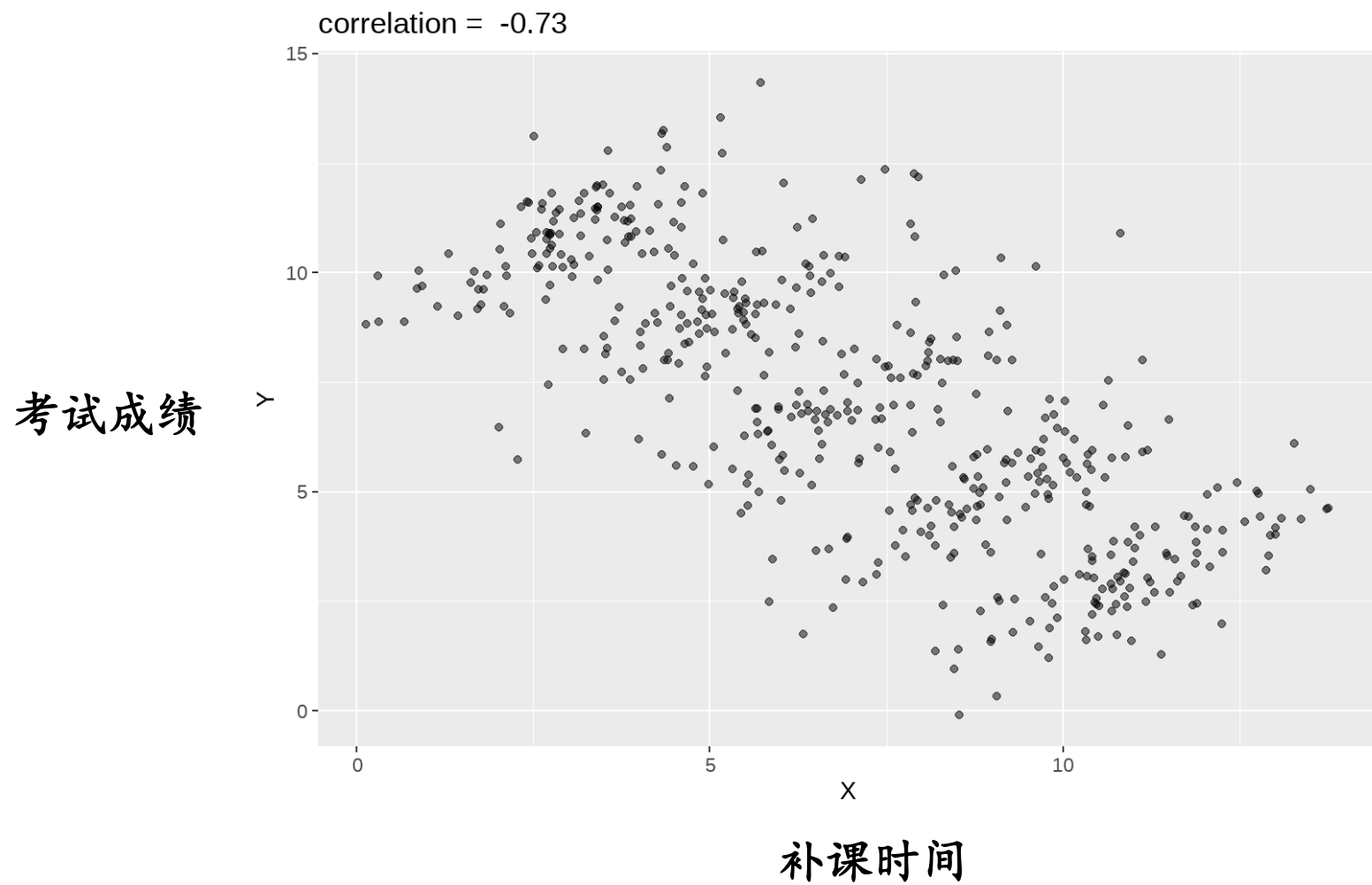
资源下载

- [diamonds.csv \(第 1 课\)](#)
- [wages.csv \(第 4 课\)](#)
- [完整示例代码 visualization.R](#)
- [图 1: 图形属性示意图](#)

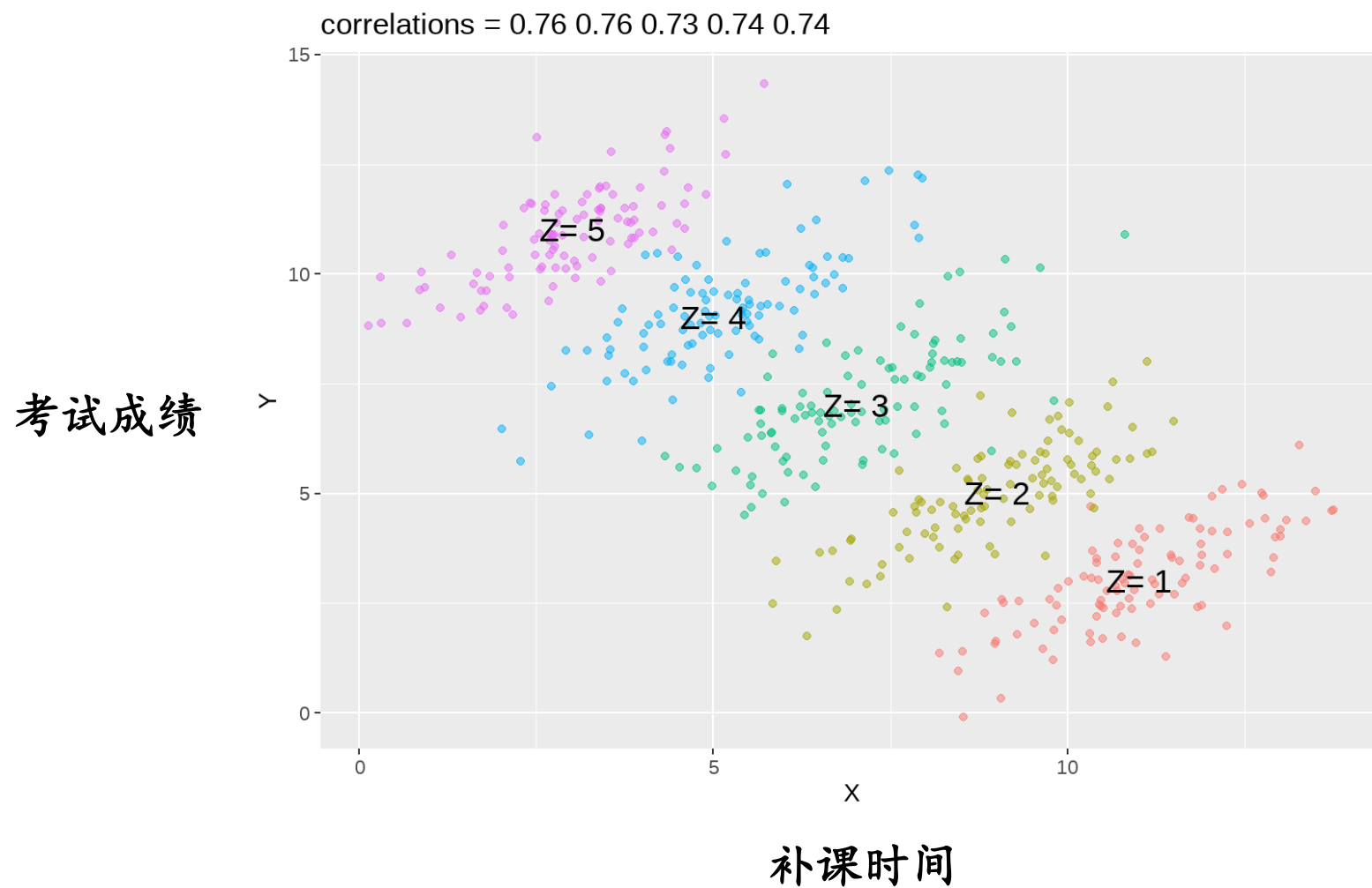
为什么要可视化？



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宏包ggplot2

```
> library(cranlogs)
>
> d <- cran_downloads(package = "ggplot2", from = "2025-01-01", to = "2025-10-31")

Warning message:
In fill_in_dates(df, as.Date(res1$start), as.Date(res1$end)) :
  Time interval in the future

>
> sum(d$count)
[1] 16717239
```

```
ggplot(data = <DATA>) +
  <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>))
```

ggplot() 函数包括9个部件：

- 数据 (data) (数据框)
- 映射 (mapping)
- 几何形状 (geom)
- 统计变换 (stats)
- 标度 (scale)
- 坐标系 (coord)
- 分面 (facet)
- 主题 (theme)
- 存储和输出 (output)

Let's try some codes!

Thank you!

For more information:

- <https://chenziqu0506-ai.github.io/ziqichenRsimpletutorial/index.html>
- <https://github.com/chenziqu0506-AI/ziqichenRsimpletutorial/tree/main>